Summary in Graph

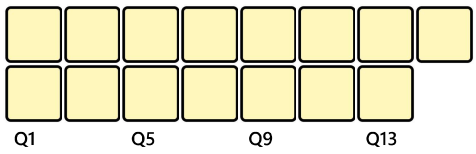
Exam Summary (GO Classes DA Test Series 2026 | Data Structures and Algorithms | Topic Wise Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	15 (5 + 10)
Total Marks:	25 (5 + 20)
Exam Duration:	45 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (47) out of 465 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

What is the time complexity of code given?

```
    def fun(n):
        count = 0
        i = n
        while i>0 :
5.          for j in range(i):
                count += 1
            i //= 2
        return count
```

A. $\Theta(\log n)$
B. $\Theta(n)$
C. $\Theta(n \log n)$
D. $\Theta\left(n^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 04sec | Discuss

Q #2 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Let $b > 1$. Then $\log_b\left(\left(n^2\right)!\right)$ is _____

A. $\Theta\left(\log_b(n!)\right)$
B. $\Theta\left(\log_b(2n!)\right)$
C. $\Theta\left(n \log_b(n)\right)$
D. $\Theta\left(n^2 \log_b(n)\right)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Select Type | Award: 1 | Penalty: 0 | Algorithms

Which of the following(s) is/are true about positive functions

A. $f(n) = \Omega(g(n)) \implies g(n) = O(f(n))$
B. $f(n) = \omega(g(n)) \implies f(n) = \Omega(g(n))$
C. $f(n) = O(g(n)) \wedge f(n) = \Omega(h(n)) \implies g(n) = \Theta(h(n))$
D. $o(g(n)) \cap \omega(g(n)) = \emptyset$

Your Answer: | Correct Answer: A;B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Arrange the following functions in their increasing order of growth. In options $\log^2 n$ means $(\log n)^2$

A. $\log(n!), \log^2 n, (\lg n)!, e^n$
B. $\log(n!), \log^2 n, e^n, (\lg n)!$
C. $\log^2 n, \log(n!), (\lg n)!, e^n$
D. $\log^2 n, (\lg n)!, \log(n!), e^n$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Algorithms

Consider the function $f(N) = \sum_{i=0}^{N} \sum_{j=0}^{i} j$

Which of the following is true about $f(N)$?

A. $f(N) = \Theta\left(N^2\right)$
B. $f(N) = \Theta\left(N^3\right)$
C. $f(N) = \Theta(N)$
D. $f(N) = \Theta(N \log N)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider Iterated logarithm of n, written $\log^* n$ (usually read "log star n"), is the number of times the logarithm (base 2) function must be iteratively applied before the result is less than or equal to 1. Mathematically,

$$\lg^* N = \min\{k \mid \underbrace{\lg \lg \cdots \lg}_{k} N \leq 1\}$$

For example-

$$\lg^* 2 = 1,$$
$$\lg^* 4 = 2,$$
$$\lg^* 16 = 3,$$
$$\lg^* 65536 = 4,$$
$$\lg^*\left(2^{65536}\right) = 5$$

Which of the following is/are correct about the implementation of $\log * n$?

In all options, $\log()$ is a function which calculates $\log$ base 2 of a given number.

A.
```
import math
def iterated_log(n):
    if n > 1.0:
        return 1 + iterated_log(math.log2(n))
    else:
        return 0
```

B.
```
import math
def iterated_log(n):
    count = 0
    while n > 1.0:
        n = math.log2(n)
        count += 1
    return count
```

C. $\log * n = \begin{cases} 0 & \text{if} \quad n \leq 1 \\ 1 + \log * (\log n) & \text{if} \quad n > 1 \end{cases}$

D. $\log^* 2^m = \begin{cases} 1 + \log^* m & \text{if} \quad m > 1 \\ 1 & \text{if} \quad m \leq 1 \end{cases}$

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the following C function:

```
    def fun1(n):
        q = 0
        for i in range(1, n):
            p = 0
 5.         j = n
            while j > 1:
                p += 1
                j = j // 2
            k = 1
10.         while k < p:
                q += 1
                k = k * 2
        return q
```

Which one of the following most closely approximates the return value of the function fun1?

A. n^3
B. $n(\log n)^2$
C. $n \log n$
D. $n \log(\log n)$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

What will be the time complexity of following code?

```
    def mystery(N):
        i = 1
        s = 1
        while s <= N:
 5.         i += 1
            s = s + i
```

A. $\Theta(\sqrt{N})$
B. $\Theta(N)$
C. $\Theta(\log N)$
D. $\Theta\left((\log N)^2\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

```
    def fun(N):
        i = N
        while i > 0:
            for j in range(i * 2):
 5.             print("Hello World")
            i = i // 2
```

A. $\Theta(N \log N)$
B. $\Theta(N)$
C. $\Theta(\log N)$
D. $\Theta\left((\log N)^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

What is the time complexity of following code?

```
    def fun(n):
        count = 0
        i = n
        while i > 0:
5.          for j in range(i):
                count += 1
            i //= 2
        return count
```

A. $\Theta(N \log N)$
B. $\Theta(N)$
C. $\Theta(\log N)$
D. $\Theta\left(N^2\right)$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider the case: $f(n) = O(g(n))$. Then, following two statements are claimed to be inferred from the above case.

- $\mathrm{S1}: 2^{f(n)} = O\left(2^{g(n)}\right)$
- $\mathrm{S2}: 2^{g(n)} = O\left(2^{f(n)}\right)$

Choose the correct option from the given.

A. Both correct.
B. Both incorrect
C. $\mathrm{S1}$ correct, $\mathrm{S2}$ false
D. $\mathrm{S2}$ false, $\mathrm{S1}$ correct

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Which of the following(s) is/are TRUE?

A. $\sum_{i=0}^{n} 5^{-i} = \Theta(n)$

B. $2^{\log_4 n} = \Theta(\sqrt{n})$

C. $\sum_{i=1}^{\log m} c2^{2i} = O\left(4^{\log m}\right)$

D. $\sum_{i=1}^{n^2} \mathrm{i} \log \mathrm{n} = \Omega\left(\mathrm{n}^3\right)$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Suppose we perform a sequence of n operations on a data structure in which the i th operation costs i if i is an exact power of 2, and 1 otherwise.
Let C_i be cost of i th operation then -

$$C_i = \begin{cases} i & \text{if i is an exact power of 2} \\ 1 & \text{otherwise} \end{cases}$$

What will be the total cost of n operations?

A. $\Theta(n)$
B. $\Theta(n \log n)$
C. $\Theta\left(2^n\right)$
D. $\Theta\left(n2^n\right)$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Select Type | Award: 2 | Penalty: 0 | Algorithms

Consider a set of functions in $2^{(O(n))}$ which includes all functions of form $2^{(s(n))}$ where $s(n) = O(n)$.

For example :- $2^n = 2^{(O(n))}, 4^n = 2^{(O(n))}$ and so on.

Which of the following(s) is/are FALSE?

A. $f(n) = \left(\frac{n}{2}\right)^n$ then $f(n) = 2^{\Theta(n \log n)}$
B. $3^n = O\left(2^n\right)$
C. $3^n = 2^{O(n)}$
D. $\frac{\ln n}{\sqrt{n}}$ is $O(\sqrt{n})$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Algorithms

Consider functions $f(), g(), \text{and } h()$ which are positive increasing functions.

Also, consider two statements below -

S1: If $f(n) = o(g(n))$ then $h(f(n)) = o(h(g(n))$, here o is small-oh

S2: If $f(n) = O(g(n))$ then $h(f(n)) = O(h(g(n))$, here O is big-oh

Which of the following is correct ?

A. S1 is true but S2 is false
B. S1 is false but S2 is true
C. Both statements are true
D. Both Statements are false

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

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